Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/17/2024
PI Name:	Allaway, Heather Ph.D.		
Project Title:	Insights Into the Impacts of Continuous, Low Dose-Rate Neutron Radiation Exposure on Maternal and Fetal Skeletal Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	hallaway@lsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-431-8318
Organization Name:	Louisiana State University and A&M College		
PI Address 1:	Kinesiology		
PI Address 2:	2139 Huey P Long Field House		
PI Web Page:			
City:	Baton Rouge	**State:**	LA
Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	05/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 5/22/24).		
Key Personnel Changes/Previous PI:	No Key Personnel Changes were made during the reporting period We lost an unpaid collaborator and developed a new unpaid collaboration.		
COI Name (Institution):	Lau, Anthony Ph.D. (College of New Jersey)		
Grant/Contract No.:	80NSSC23K0784		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The combined effects of space environmental stressors induce pathologies in multiple organ systems. Specifically, losses to the musculoskeletal system may be very dangerous for the health and performance of astronauts on extended duration missions on the Moon or arriving on Mars. A critical need remains to better understand the impact of radiation exposure, one of the key environmental stressors of deep space, on human and animal physiology to enable extended duration missions beyond low Earth orbit or setting up settlements on the Moon or Mars. There is a critical gap in knowledge surrounding the impact of the space radiation environment on skeletal health and on the progress of fetal skeletal development during pregnancy. The objective of the current proposal is to capitalize on a unique tissue-sharing opportunity to examine the combined effects of continuous radiation exposure and pregnancy on maternal and fetal skeletal physiology. We propose to assess maternal and fetal skeletal physiology through measurements of mineral and material properties, as well as assess changes in cellular dynamics of the maternal bone under the microscope. This study will be critical in assessing how a very harmful component of the space radiation environment impacts multiple aspects of skeletal health, including sex-specific differences and individual variation in the impact of the space environment on the functioning of the body.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides new insights on how near continuous low-dose of ionizing radiation (<0.1Gy) affects the maternal fetal system related to bone. The maternal and fetal skeletal changes from these extremely low doses could be considered exposure to clinical radiation for diagnostic and radiotherapies.
Task Progress:	During the past reporting period, the external morphology assessments of the fetal samples (AIM 2) were performed by undergraduate research assistants. We have taken caliper measurements of the crown rump length, head length and width, thoracic width, abdominal depth and width, tail length and width at insertion to the body, lengths of both forelimbs and forepaws, and lengths of both hindlimbs and hindpaws. While the external morphology measurements have been taken, images of all samples are also being taken using a dissecting microscope. On the E18 samples, we have also been able to complete some external facial measurements (interorbital width, nostril spacing, mouth width, and snout height). All protocols for these external morphology measurements were based on prior publications; however, the protocols were developed specifically for this project. The E12 and E18 available morphological data has been presented at local conferences, as well as the American Society for Gravitational and Space Research (ASGSR). Statistical analyses were performed using a nested t-test in GraphPad Prism. The nested t-test accounts for the genetic similarity among samples from the same litter. We are still exploring options for labs to collaborate with, or gain access to equipment, to perform the sex determination analysis on the fetal samples. Though the data gathered will still be useful without this information, the determination of fetal sex will enhance the data collected. Our original collaborator accepted another position at a university in a different state, and thus we are in search of another collaborator. We have to wait to perform this analysis until all other non-destructive data collection is complete, as this analysis requires removing a portion of the fetal tail. We conducted a proof-of-concept microCT scan of the E18 fetal chest area to garner what soft tissue information we could observe without any staining, and to determine the best settings for getting bone data for these samples using our microcomputed tomography (microCT) machine. During ASGSR, we began developing an additional collaboration to increase the data that could be gathered with the microCT scanning of the fetal samples. The scanner available to a potential collaborator can have an attenuator sheet placed between the sample and the beam, which would enable scanning of bones and soft tissue without needing to stain the samples or perform any other destructive measurements. This is not possible with the core equipment available at the Principal Investigator's (PI's) institution. We are in the process of preparing a few E18 and E12 samples to send to our new collaborator for a test scan before sending all of the samples in batches. MicroCT scanning of the maternal tibia (AIM 1) has been completed at the proximal metaphyseal area and midshaft. Our team is actively working on analyses of the scans. A few bones may need to be rescanned, which currently prevents us from shipping the samples to the Co-Investigator (Co-I) for mechanical testing. MicroCT scans of the femur remain to be completed, but time on the microCT machine is currently at a premium. Due to the potential need to rescan samples, we have nothing to report regarding the mechanical testing planned. Plastic embedding of the samples and the re-embedding is complete. Training of a new student is underway to complete the histologic sectioning. As the samples are from a mouse, we are being very careful to ensure that no sample is lost due to lack of training. Samples for paraffin embedding have been decalcified and will be taken to the histology core for embedding in May 2024. After embedding in paraffin, we will begin the immunohistochemical and marrow adipose staining and analyses.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	LoGiudice M, Steller J, Sorrels C, Chancellor J, Allaway H. "Impacts of 18 days of continuous neutron irradiation on fetal development." 10th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 21, 2023. Abstracts. 10th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 21, 2023. , Apr-2023
Abstracts for Journals and Proceedings	Sorrels C, Steller J, LoGiudice M, Chancellor J, Allaway H. "Impacts of 12 days of continuous neutron irradiation on fetal development." 10th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 21, 2023. Abstracts. 10th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 21, 2023. , Apr-2023
Abstracts for Journals and Proceedings	LoGiudice M, Steller J, Sorrels C, Chancellor J, Allaway H. "Impact of continuous, low-dose-rate neutron irradiation during gestation on fetal development." American Society for Gravitational and Space Research Annual Conference, Washington DC, November 12-18, 2023. Abstracts. American Society for Gravitational and Space Research Annual Conference, Washington DC, November 12-18, 2023. , Nov-2023

Abstracts for Journals and Proceedings	Beard C, Steller J, Chancellor J, Lau A, Allaway H. "Impacts of continuous, low-dose-rate neutron radiation exposure on maternal skeletal physiology." Geauxing the Distance with Kinesiology (inaugural research symposium), LSU School of Kinesiology, Baton Rouge, LA, April 4-5, 2024. Abstracts. Geauxing the Distance with Kinesiology (inaugural research symposium), LSU School of Kinesiology, Baton Rouge, LA, April 4-5, 2024. , Apr-2024
Abstracts for Journals and Proceedings	LoGiudice M, Steller J, Sorrels C, Chancellor J, Allaway H. "The effects of continuous, low-dose neutron radiation exposure during gestation on the external morphology and development of the mouse fetus at day 18 of development." Geauxing the Distance with Kinesiology (inaugural research symposium), LSU School of Kinesiology, Baton Rouge, LA, April 4-5, 2024. Abstracts. Geauxing the Distance with Kinesiology (inaugural research symposium), LSU School of Kinesiology, Baton Rouge, LA, April 4-5, 2024. , Apr-2024
Abstracts for Journals and Proceedings	LoGiudice M, Sorrels C, Steller J, Chancellor J, Allaway H. "Impact of continuous, low-dose-rate neutron irradiation during gestation on fetal development." 11th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 19, 2024. Abstracts. 11th Annual LSU Discover Day, Louisiana State University Undergraduate Research Symposium, Baton Rouge, Louisiana, April 19, 2024. , Apr-2024
Awards	LoGiudice M, Sorrels C. "LSU Discover Day 2024, Second Place Poster in Science, Technology, Engineering, and Mathematics, April 2024." Apr-2024
Awards	LoGiudice M, Sorrels C. "LSU Discover Day 2024, College of Human Sciences and Education Second Place Poster/Presentation, April 2024." Apr-2024
Awards	LoGiudice M. "LSU Discover Day 2023, College of Human Sciences and Education First Place Poster/Presentation, April 2023." Apr-2023
Awards	LoGiudice M. "LSU Discover Day 2023, Second Place Poster in Science, Technology, Engineering, and Mathematics, April 2023." Apr-2023
Awards	LoGiudice M. "2023-2024 Tiger Athletic Foundation Honors College Thesis Research Scholarship, December 2023." Dec-2023
Awards	LoGiudice M. "LSU Discover Undergraduate Research Program Travel Award, October 2023." Oct-2023
Awards	LoGiudice M. "LSU Organizations Relief Fund for Undergraduate Travel Award, October 2023." Oct-2023
Awards	Allaway H. "College of Human Sciences & Education Early Career Award, Louisiana State University, January 2024." Jan-2024
Awards	LoGiudice M. "Distinguished Undergraduate Researcher, Discover Undergraduate Research Program, Louisiana State University, May 2024." May-2024
Awards	LoGiudice M. "The Leo & Insa Abraham Honors Excellence Award in Honor of Dr. James D. Hardy Jr., Ogden Honors College, Louisiana State University, May 2024." May-2024
Dissertations and Theses	LoGiudice M. "The effects of continuous, low-dose neutron radiation exposure during gestation on the external morphology and development of the mouse fetus at day 18 of development." Undergraduate Honors Thesis, Louisiana State University, May 2024. , May-2024